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ARTICLE

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Published: 01 September 1980

[Citation in BibTeX format](#)

ALGORITHM 560

JNF, An Algorithm for Numerical Computation of the Jordan Normal Form of a Complex Matrix [F2]

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Key Words and Phrases Jordan normal form, canonical form, eigenvalues, eigenvectors, principal vectors, block diagonal form

CR Categories. 5 14

Language. Fortran

DESCRIPTION

1. Introduction

The routines given here are the actual Fortran implementation of the algorithm presented and discussed in [2]. We describe in detail how to use the Fortran subroutines and how to reach the results from a call. We also give some comments on the code that might be of value when implementing the subroutines on a particular machine. The subroutines have been checked with the PFORT verifier [4] and the notation from [2] is used.

2. The User-Written Routine DECIDE

DECIDE is a user-written subroutine which makes it possible for the user to change the grouping and/or the values of the numerical multiple eigenvalues.

This routine is useful when we have some information on the eigenvalues and their multiplicities in advance, and want the eigenvectors and the principal vectors for the given matrix. For instance, if we know of physical reasons why zero should be the only possible multiple eigenvalue and all others simple, then the contents of the parameters should be changed as indicated in Table I.

If the user does not want to influence the grouping or the values of the eigenvalues, it is enough to define a dummy subroutine, i.e.,

```
SUBROUTINE DECIDE (NM, N, NDEL, CSHTR, CSHTI, NBLOCK, HR, HI)
RETURN
END
```

Received 21 May 1975 and August 1977.

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This work was supported by the Swedish Natural Science Research Council.

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Table I. Example of Use of the Subroutine DECIDE

On entry N = 10 NBLOCK = 6		On exit NBLOCK = 7	
NDEL	CSHT	NDEL	CSHT
0	↗ -1.5	0	↗ -1.5
1	↘ 0.002	1	↘ 0.0
5	2.4	5	<u>H(6, 6)</u>
7	3.5	6	<u>H(7, 7)</u>
8	8.6	7	3.5
9	↗ 9.5	8	8.6
10	↘	9	9.5
		10	

3. The Rank Determination Process and the RDEFL Routine

The rank determination process is described in [2, Section 2.2 and Section 3, Step 6 of the algorithm]. According to these two alternatives we have two subroutines RDEFL—one for the singular value decomposition strategy and one for the Kublanovskaya (RQ) decomposition strategy. Only the routine based on singular value decomposition is presented here.

In the RDEFL routine we use the CSVD algorithm [1] with one change; namely, we sort the singular values in increasing instead of decreasing order as in the original program.

4. How to Reach Any of the Vectors

The eigenvectors and principal vectors are stored in the arrays ZR, ZI(NM, N). To find the vectors, we use the information stored in NBLOCK, NDEL, NDB, NDEFL, and NXT (see the parameter list of subroutine JNF).

The column indices K in ZR and ZI for eigenvectors, principal vectors, and principal chains of an eigenvalue are listed below in Algol for-statement notation.

- (a) Find the eigenvectors of the multiple eigenvalue
 $I = 1, \dots, \text{NBLOCK}.$
 $J := \text{NDB}[I];$
for $K := \text{NDEL}[I] + 1$ **step** 1 **until** $\text{NDEFL}[J + 1] - 1$ **do**;
- (b) Find the principal vectors of grade P to the eigenvalue $I = 1, \dots, \text{NBLOCK}.$
 $J := \text{NDB}[I] + P - 1;$
for $K := \text{NDEFL}[J]$ **step** 1 **until** $\text{NDEFL}[J + 1] - 1$ **do**;
- (c) Find the principal chain which ends with the eigenvector in column L.
 First:
 $(A - \text{EV}[L] * I) * Z[L] = 0$ (eigenvector)
 Then:
 $K1 := L;$
for $K := \text{NXT}[K1]$ **while** $K > 0$ **do**
 begin
 Here:
 $(A - \text{EV}[K] * I) * Z[K] = \text{SUPD}[K1] * Z[K1];$ (principal vector)
 $K1 := K$
 end;

Table II. Example of Output of Index Vectors

I	NDEL	NDB	NDEFL	NXT	SUPD
1	0	1	1	3	1.9903
2	4	3	3	4	48.080
3	9	6	5	0	0
4	10	0	7	0	0
5	0	0	9	7	3.1314
6	0	0	10	8	5.5354
7	0	0	11	0	0
8	0	0	0	9	1.3994
9	0	0	0	0	0
10	0	0	0	0	0

Note. Output for the first test matrix in [2] is displayed.

(Here we use the fact that the diagonal elements may be different as in [2, eq. (4.2)]. To get chains of the nilpotent B, EV[K] should be replaced by EV[L].)

In Table II we list the values of NDEL, NDEFL, NXT, NDB, and SUPD for the first test matrix in [2, Section 5].

5. Comments on the Fortran Code

Since we use the EISPACK routines CBAL, COMHES, COMLR2, and CBABK2 [5], we have their representation of complex arrays, i.e., the real parts are represented in one array with the name ending in R (e.g., HR, ZR) and the imaginary parts are represented in one array with the name ending in I (e.g., HI, ZI). This representation makes it possible to execute most of the computations in real arithmetic.

We use real arithmetic in all steps of the algorithm except Steps 4 and 7, i.e., the elimination processes, where we use some complex arithmetic.

Because of this complex arithmetic we also have to use the complex standard functions REAL(X), AIMAG(X), and CMPLX(X, Y).

If the actual Fortran compiler does not accept complex arithmetic, we can implement the functions CMUL, CDIV, CADD, and CSUB from [6] and substitute the actual statements with new statements.

A new parameter (NOBACK) has been added to the EISPACK routine COMLR2. We use this parameter (NOBACK = 0) as a control so that we do not get the eigenvectors from COMLR2. We get the transformations which transform the matrix to upper triangular form.

In the RDEFL routine we use complex arrays and arithmetic since the routine CSVD1 [1] is made for complex arithmetic.

6. Numerical Experiments

We have performed several numerical tests, both on matrices with a well-defined Jordan normal form, and on cases specially constructed in order to provide the routine with difficulties. Results from full Jordan normal form reduction are reported and discussed in [2]. In [2, Section 4] we describe how to choose tolerance parameters in the grouping procedure and in the procedure for determining the structures inside the invariant subspaces corresponding to different

numerical multiple eigenvalues. We also describe how to analyze the results from the program, given a combination of the tolerance parameters.

A large number of results from block diagonal reductions (ISTEP = 5 and 6) are reported and discussed in [3].

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ALGORITHM

[Summary information and a part of the listing are printed here. The complete listing is available from the ACM Algorithms Distribution Service (see inside back cover for order form) or may be found in "Collected Algorithms from ACM."]

NAME(*n*): indicates a Fortran module with *n* records

NAME^T(*n*): indicates "NAME" is included for testing purposes

Contents: MAIN^T(328), RESID^T(75), DECIDE^T(56), MATRIS^T(43),
MATRS1^T(32), MATRS3^T(55), MATRS4^T(62), MATRS5^T(42),
JNF(834), RDEFL(132), CBAL(209), COMHES(138),
COMLR2(405), CBABK2(91), CVSD1(384), OUTPUT^T(33),
MATBLK^T(37), MATRS2^T(148)

```

      SUBROUTINE DECIDE (NM,N,NDEL,CSHTR,CSHTI,NBLOCK,HR,HI)
C *****
C *
C *  DECIDE IS A USER WRITTEN SUBROUTINE WHICH MAKES IT POSSIBLE
C *  FOR THE USER TO CHANGE THE GROUPING OF THE NUMERICAL MUL-
C *  TIPLE EIGENVALUES,AND/OR CHANGE THE VALUES OF THE MULTIPLE
C *  EIGENVALUES.
C *  THE FORMAL PARAMETER LIST
C *
C *  NM      THE ROW DIMENSION OF THE TWO-DIMENSIONAL ARRAY
C *          PARAMETERS(HR,HI)AS DECLARED IN THE CALLING
C *          PROGRAM DIMENSION STATEMENT
C *  N       THE ORDER OF THE ORIGINAL MATRIX
C *  NBLOCK  THE NUMBER OF BLOCKS I.E THE NUMBER OF NUMERICAL
C *          MULTIPLE EIGENVALUES
C *  CSHTR,  THE REAL AND IMAGINARY PARTS,RESPECTIVELY,OF THE
C *  CSHTI   NUMERICAL MULTIPLE EIGENVALUES
C *  NDEL    INDICATES THE MULTIPLICITY OF THE NUMERICAL
C *          MULTIPLE EIGENVALUES.NDEL(I+1)-NDEL(I)=THE
C *          MULTIPLICITY OF EIGENVALUE I FOR I=1,...,NBLOCK

```

```

C * HR,HI THE REAL AND IMAGINARY PARTS,RESPECTIVELY, OF 00004240
C * THE UPPER TRIANGULAR MATRIX RESULTING FROM 00004250
C * STEPS 1-3 OF THE ALGORITHM 00004260
C * 00004270
C * NOTE -- THE USER MUST NOT CHANGE HR,HI 00004280
C * 00004290
C ***** 00004300
C INTEGER NDEL(N) 00004310
C REAL HR(NM,N),HI(NM,N),CSHTR(1),CSHTI(1) 00004320
C WRITE(6,9) 00004330
9 FORMAT(1H0,1X,47HTHE FOLLOWING OUTPUT (A,B AND C) ARE PRINTED BY, 00004340
* 31HTHE USER WRITTEN ROUTINE DECIDE/ 00004350
* 1H0,1X,31HSEE SECTION 2 OF THE ALGORITHM.) 00004360
C WRITE(6,10) 00004370
C WRITE(6,11) (HR(K,K),K=1,N) 00004380
C WRITE(6,11) (HI(K,K),K=1,N) 00004390
C WRITE(6,12) 00004400
C DO 1 K=1,NBLOCK 00004410
C MULT=NDEL(K+1)-NDEL(K) 00004420
C WRITE(6,13) NDEL(K+1),MULT,CSHTR(K),CSHTI(K) 00004430
1 CONTINUE 00004440
C WRITE(6,14) 00004450
14 FORMAT(1H0,1X,34HC--IN STEP 6 OF THE ALGORITHM THE , 00004460
* 50HSTRUCTURE OF EACH MULTIPLE EIGENVALUE IS COMPUTED. 00004470
* /1H0,1X,42HFOR THAT REASON RDEFL SUCCESIVELY COMPUTES, 00004480
* 44H SINGULAR VALUE DECOMPOSITIONS. RDEFL PRINTS 00004490
* /1H0,1X,39HTHE RESULTS BELOW(SEE ALSO COMMENTS IN , 00004500
* 35HRDEFL AND STEP 6 OF THE ALGORITHM.) 00004510
10 FORMAT(1H0,1X,43HA--ENTER DECIDE WITH EIGENVALUES COMPUTED , 00004520
* 36HBY COMLR2 (STEP 1 OF THE ALGORITHM )) 00004530
11 FORMAT(10F13.9) 00004540
12 FORMAT(1X,40HB--GROUPINGS OF THE EIGENVALUES,COMPUTED, 00004550
* 27H BY STEP 3 OF THE ALGORITHM) 00004560
13 FORMAT(12H DIVISION AT,I4, 7H MULT.=,I4, 8H CENTER=,2E20.10) 00004570
C RETURN 00004580
C END 00004590

C SUBROUTINE JNF (NM,N,HR,HI,ZR,ZI,EVR,EVI,SUPD,NXT,NDEL,NDEFL,NDB, 00006940
C CNBLOCK,EIN,TOL,DELE,SM,IERR,ISTEP) 00006950
C REAL HR(NM,N),HI(NM,N),ZR(NM,N),ZI(NM,N),EVR(N),EVI(N),SUPD(N), 00006960
C CDELE(N),SM(N) 00006970
C INTEGER NXT(N),NDEL(N),NDEFL(N),NDB(N) 00006980
C 00006990
C ***** 00007000
C * THE FORMAL PARAMETER LIST 00007010
C * 00007020
C * ON INPUT - 00007030
C * 00007040
C * NM - MUST BE SET TO THE ROW DIMENSION OF THE TWO- 00007050
C * DIMENSIONAL ARRAY PARAMETERS AS DECLARED IN 00007060
C * THE CALLING PROGRAM 00007070
C * N - IS THE ORDER OF THE MATRICES 00007080
C * 00007090
C * HR,HI - CONTAIN THE REAL AND IMAGINARY PARTS,RESPECTIVELY 00007100
C * OF THE COMPLEX MATRIX 00007110
C * EIN - IS A TOLERANCE PARAMETER,CORRESPONDING TO 00007120
C * PERTURBATIONS OF HR,HI, AND IS USED IN THE 00007130
C * GROUPING OF NUMERICAL MULTIPLE EIGENVALUES 00007140
C * 00007150
C * TOL - IS A TOLERANCE PARAMETER USED IN THE CONSTRUCTION 00007160
C * OF THE NILPOTENT MATRICES (IS USED IN RDEFL) 00007170
C * 00007180

```

```

C *      ISTEP - RETURN FROM JNF AFTER STEP ISTEP(=1,2,3,4,5,6,7) 00007190
C *      FULL REDUCTION TO JORDAN FORM IF ISTEP .GE. 7 00007200
C *      OTHER CHOICES GIVE REDUCTION TO 00007210
C *      00007220
C *      1      UPPER TRIANGULAR FORM 00007230
C *      00007240
C *      2      UPPER TRIANGULAR FORM WITH THE EIGEN- 00007250
C *      VALUES SORTED SUCH THAT CLOSE EIGEN- 00007260
C *      VALUES APPEAR IN ADJACENT POSITIONS 00007270
C *      00007280
C *      3      THE SAME AS 2, IN ADDITION GROUPING OF 00007290
C *      CLOSE EIGENVALUES INTO BLOCKS CORRE- 00007300
C *      SPONDING TO NUMERICAL MULTIPLE EIGEN- 00007310
C *      VALUES 00007320
C *      00007330
C *      4      BLOCK DIAGONAL UPPER TRIANGULAR FORM 00007340
C *      00007350
C *      5      BLOCK DIAGONAL UPPER TRIANGULAR FORM 00007360
C *      AND THE INVARIANT SUBSPACES CORRESPONDING 00007370
C *      TO THE DIAGONALS BLOCKS HAVE UNITARY BASES 00007380
C *      00007390
C *      6      THE STRUCTURE OF EACH DIAGONAL BLOCK IS 00007400
C *      DETERMINED 00007410
C *      ON OUTPUT - 00007420
C *      00007430
C *      THE PARAMETERS CONTAIN USEFUL INFORMATION DEPENDING ON 00007440
C *      THE VALUE OF THE INPUT PARAMETER ISTEP. 00007450
C *      00007460
C *      NOTATION - (IF ISTEP .GE. N ) 00007470
C *      WHERE N IS 1,2,3,4,5,6 OR 7 00007480
C *      00007490
C *      00007500
C *      HR,HI - HAVE BEEN DESTROYED 00007510
C *      00007520
C *      IERR - IS A CONVERGENCE PARAMETER SET BY COMLR2 00007530
C *      0 FOR NORMAL RETURN 00007540
C *      J IF THE J-TH EIGENVALUE HAS NOT BEEN DETERMINED 00007550
C *      AFTER 30 ITERATIONS 00007560
C *      IF AN ERROR EXIT IS MADE (IERR.NE.0),NONE 00007570
C *      OF THE FOLLOWING PARAMETERS CONTAIN MEANINGFUL 00007580
C *      RESULTS 00007590
C *      (IF ISTEP.GE.1) 00007600
C *      EVR,EVI CONTAIN THE REAL AND IMAGINARY PARTS,RESPECTIVELY 00007610
C *      OF THE EIGENVALUES. 00007620
C *      IF ISTEP.GE.1 BUT .LT.5 THE EIGENVALUES ARE 00007630
C *      COMPUTED BY COMLR2. 00007640
C *      IF ISTEP.GE.6, EVR,EVI CONTAIN THE MAIN DIAGONAL 00007650
C *      OF THE JORDAN MATRIX. 00007660
C *      00007670
C *      NBLOCK- IS THE NUMBER OF NUMERICAL MULTIPLE EIGENVALUES 00007680
C *      (IF ISTEP.GE.3) 00007690
C *      00007700
C *      NDEL - INDICATES THE MULTIPLICITIES OF THE NUMERICAL 00007710
C *      MULTIPLE EIGENVALUES 00007720
C *      (IF ISTEP.GE.3) 00007730
C *      NDEL(I+1)-NDEL(I)= MULTIPLICITY OF EIGENVALUE 00007740
C *      I FOR I=1,...,NBLOCK 00007750
C *      NDEFL - INDICATES THE STRUCTURE OF HR,HI 00007760
C *      (IF ISTEP.GE.6) 00007770
C *      00007780
C *      SUPD - CONTAINS THE REAL SUPERDIAGONAL OF THE JORDAN 00007790
C *      MATRIX (THE COUPLING ELEMENTS) 00007800
C *      (IF ISTEP.GE.7) 00007810

```

```

C  *
C  *      NXT  - CONTAINS THE COLUMN INDICES OF SUPD      00007820
C  *      NXT(I)=J IMPLIES THAT THE VECTOR WITH COUPLING  00007830
C  *      ELEMENT SUPD(I) IS PLACED IN COLUMN             00007840
C  *      J OF ZR,ZI                                     00007850
C  *      (IF ISTEP.GE.7)                                00007860
C  *      NDB  - CONTAINS POINTER REFERENCES              00007870
C  *      NDB(I)=J MEANS THAT INFORMATION ABOUT THE      00007880
C  *      STRUCTURE OF THE NUMERICAL MULTIPLE           00007890
C  *      EIGENVALUE I STARTS IN POSITION J               00007900
C  *      OF NDEFL                                       00007910
C  *      (IF ISTEP.GE.6)                                00007920
C  *      ZR,ZI - CONTAIN THE REAL AND IMAGINARY PARTS,RESPECTIVELY 00007930
C  *      OF THE ACCUMULATED TRANSFORMATIONS FROM STEP 1 00007940
C  *      TO 7 OF THE ALGORITHM.                        00007950
C  *      IF ISTEP.GE.7, ZR,ZI ARE THE EIGENVECTORS AND  00007960
C  *      THE PRINCIPAL VECTORS.                        00007970
C  *                                                    00007980
C  *      DELE - CONTAINS INFORMATION ABOUT DELETED ELEMENTS 00007990
C  *      DURING THE PROCESSES OF FINDING NILPOTENT MATRICES 00008000
C  *      DELE(I)= EUCLIDEAN NORM OF DELETED PART FOR BLOCK 00008010
C  *      I FOR I=1,...,NBLOCK                          00008020
C  *      (IF ISTEP.GE.6)                                00008030
C  *                                                    00008040
C  *                                                    00008050
C  *      SM   - CONTAINS ESTIMATES OF THE SPECTRAL PROJECTORS 00008060
C  *      CORRESPONDING TO DIFFERENT NUMERICAL MULTIPLE  00008070
C  *      EIGENVALUES                                   00008080
C  *      (IF ISTEP.GE.4)                                00008090
C  *                                                    00008100
C  *      FOR COMPLETE DESCRIPTION SEE COMPUTATIONAL DETAILS IN 00008110
C  *      THE TEXT                                       00008120
C  *                                                    00008130

```
